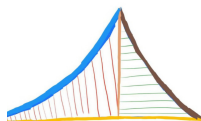


Math for EE

1



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ABOUT THIS BOOK

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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First manual appeared in January 2018

First edition published on July 10, 2024

In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

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1. GATE AE 2026

A. Formulae

1.1.1 Probability

- Addition Theorem of Probability:

$$P(M) + P(E) - P(ME) + P((M + E)') = 1 \quad (1.1.1.1)$$

- Complement Laws in Boolean algebra

$$A + A' = 1, \quad AA' = 0 \quad (1.1.1.2)$$

- Probability of Certain and Impossible Events

$$P(1) = 1, \quad P(0) = 0 \implies P(A + A') = 1, \quad P(AA') = 0 \quad (1.1.1.3)$$

- Addition Law of Probability for Mutually Exclusive Events

$$P(A + A') = P(A) + P(A') \quad (1.1.1.4)$$

- Difference rule of probability

$$P(AB') = P(A) - P(AB) \quad (1.1.1.5)$$

Solution: From (1.1.1.2)

$$A = A \cdot 1 = A(B + B') \quad (1.1.1.6)$$

Applying the distributive law:

$$A = AB + AB' \quad (1.1.1.7)$$

To apply the additive property of probability, we check the intersection of AB and AB' :

$$(AB)(AB') = A \cdot A \cdot B \cdot B' = A \cdot (BB') \quad (1.1.1.8)$$

$$= A \cdot (0) \quad (1.1.1.9)$$

$$(AB)(AB') = A \cdot 0 = 0 \quad (1.1.1.10)$$

Thus, the events AB and AB' are mutually exclusive. From (1.1.1.4)

$$P(A) = P(AB + AB') = P(AB) + P(AB') \quad (1.1.1.11)$$

Rearranging, we get:

$$P(AB') = P(A) - P(AB) \quad (1.1.1.12)$$

1.1.2 Laplace transform

- Laplace transform on $g'(t)$ - Using the definition of the Laplace transform:

$$g'(t) \xleftrightarrow{\mathcal{L}} \int_0^{\infty} e^{-st} g'(t) dt \quad (1.1.2.1)$$

Applying integration by parts with:

$$u = e^{-st}, dv = g'(t) dt \quad (1.1.2.2)$$

$$g'(t) \xleftrightarrow{\mathcal{L}} [e^{-st}g(t)]_0^\infty - \int_0^\infty (-se^{-st})g(t) dt \quad (1.1.2.3)$$

$$= \left(\lim_{t \rightarrow \infty} e^{-st}g(t) - g(0) \right) + s \int_0^\infty e^{-st}g(t) dt \quad (1.1.2.4)$$

$$0 - g(0) + sG(s) \quad (1.1.2.5)$$

$$\implies g'(t) \xleftrightarrow{\mathcal{L}} sG(s) - g(0) \quad (1.1.2.6)$$

- Laplace transform on $g''(t)$ - Using the definition for $g''(t)$:

$$g''(t) \xleftrightarrow{\mathcal{L}} \int_0^\infty e^{-st}g''(t) dt \quad (1.1.2.7)$$

Applying integration by parts with:

$$u = e^{-st}, dv = g''(t) dt \quad (1.1.2.8)$$

$$g''(t) \xleftrightarrow{\mathcal{L}} [e^{-st}g'(t)]_0^\infty - \int_0^\infty (-se^{-st})g'(t) dt \quad (1.1.2.9)$$

$$= \left(\lim_{t \rightarrow \infty} e^{-st}g'(t) - g'(0) \right) + s \int_0^\infty e^{-st}g'(t) dt \quad (1.1.2.10)$$

From (1.1.2.6), we have:

$$\mathcal{L}\{g''(t)\} = 0 - g'(0) + s(sG(s) - g(0)) \quad (1.1.2.11)$$

$$\implies g''(t) \xleftrightarrow{\mathcal{L}} s^2G(s) - sg(0) - g'(0) \quad (1.1.2.12)$$

- The First Frequency Shift Theorem - It states that if :

$$f(x) \xleftrightarrow{\mathcal{L}} F(s) \quad (1.1.2.13)$$

Then,

$$e^{ax}f(x) \xleftrightarrow{\mathcal{L}} F(s-a) \quad (1.1.2.14)$$

Solution: By definition, the Laplace transform of a function $g(x)$ is:

$$g(x) \xleftrightarrow{\mathcal{L}} \int_0^\infty e^{-sx}g(x) dx \quad (1.1.2.15)$$

$$\implies e^{ax}f(x) \xleftrightarrow{\mathcal{L}} \int_0^\infty e^{-sx}(e^{ax}f(x)) dx \quad (1.1.2.16)$$

$$= \int_0^{\infty} e^{-(s-a)x} f(x) dx \quad (1.1.2.17)$$

Recall that $F(s)$ is defined as:

$$F(s) = \int_0^{\infty} e^{-sx} f(x) dx \quad (1.1.2.18)$$

Replacing s with $(s - a)$:

$$F(s - a) = \int_0^{\infty} e^{-(s-a)x} f(x) dx \quad (1.1.2.19)$$

Therefore:

$$e^{ax} f(x) \xleftrightarrow{\mathcal{L}} F(s - a) \quad (1.1.2.20)$$

- Laplace transform on x :

$$x \xleftrightarrow{\mathcal{L}} \frac{1}{s^2} \quad (1.1.2.21)$$

Solution:

$$x \xleftrightarrow{\mathcal{L}} \int_0^{\infty} x e^{-sx} dx \quad (1.1.2.22)$$

$$= \left(-\frac{x}{s} e^{-sx} \right)_0^{\infty} + \frac{1}{s} \int_0^{\infty} e^{-sx} dx \quad (1.1.2.23)$$

Here, we assume $\text{re}(s) > 0$

$$= 0 + \frac{1}{s} \left(-\frac{1}{s} e^{-sx} \right)_0^{\infty} = \frac{1}{s^2} \quad (1.1.2.24)$$

$$\implies x \xleftrightarrow{\mathcal{L}} \frac{1}{s^2} \quad (1.1.2.25)$$

- Laplace transform of the cosine and sine functions

$$e^{jax} \xleftrightarrow{\mathcal{L}} \int_0^{\infty} e^{-sx} e^{jax} dx = \int_0^{\infty} e^{-(s-j a)x} dx \quad (1.1.2.26)$$

$$= \left(-\frac{e^{-(s-j a)x}}{s - j a} \right)_0^{\infty} = \frac{1}{s - j a} \quad (\text{Note that here } \text{re}(s) > a) \quad (1.1.2.27)$$

$$= \frac{s + ja}{(s - ja)(s + ja)} = \frac{s}{s^2 + a^2} + j \frac{a}{s^2 + a^2} \quad (1.1.2.28)$$

We know that:

$$\mathcal{L}(\cos(ax) + j \sin(ax)) = \mathcal{L}(\cos(ax)) + j \mathcal{L}(\sin(ax)) : \quad (1.1.2.29)$$

Equating the real and imaginary parts, we get:

$$\cos(ax) \xleftrightarrow{\mathcal{L}} \frac{s}{s^2 + a^2} \quad (1.1.2.30)$$

$$\sin(ax) \xleftrightarrow{\mathcal{L}} \frac{a}{s^2 + a^2} \quad (1.1.2.31)$$

- Solving a differential equation of the form:

$$\frac{d^2y}{dx^2} = -\alpha^2 y, \alpha \neq 0 \quad (1.1.2.32)$$

Applying the Laplace transform on both sides, and using (1.1.2.12), we get:

$$s^2 G(s) - sy(0) - y'(0) = -\alpha^2 G(s) \quad (1.1.2.33)$$

Consider:

$$y(0) = \beta, y'(0) = \gamma \quad (1.1.2.34)$$

Rearranging, we get:

$$G(s) = \frac{s\beta + \gamma}{s^2 + \alpha^2} \quad (1.1.2.35)$$

$$G(s) = \frac{s\beta}{s^2 + \alpha^2} + \frac{\gamma}{s^2 + \alpha^2} \quad (1.1.2.36)$$

From results (1.1.2.30) and (1.1.2.31), we get:

$$G(s) = \beta \cos(\alpha x) + \frac{\gamma}{\alpha} \sin(\alpha x) \quad (1.1.2.37)$$

1.1.3 Euler's Method of Forward Differences

- First Forward Difference - Consider a differential equation:

$$\frac{dy}{dx} = f(x, y), y(x_0) = y_0 \quad (1.1.3.1)$$

The slope of the tangent m at (x_0, y_0) is given by:

$$m = f(x_0, y_0) \quad (1.1.3.2)$$

The slope of the tangent line gives the direction in which we need to move in order to trace the curve (considering we move in small steps). Hence, we can numerically compute the approximate value of a function at a given x knowing the initial conditions.

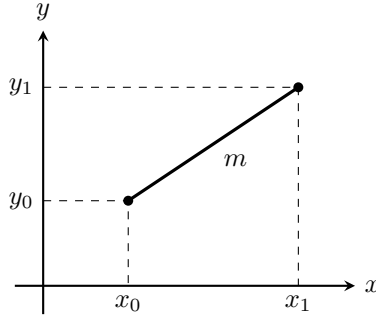


Fig. 1.1.3.1: Zoomed-in graph

Consider the above diagram, the equation of line is:

$$y_1 - y_0 = m(x_1 - x_0) \quad (1.1.3.3)$$

$$y_1 = y_0 + mh \quad (1.1.3.4)$$

Here, h is the step size(or small increment in x). Generalising the above result, we get the derivative y'_n :

$$y'_n = \frac{y_{n+1} - y_n}{h} \quad (1.1.3.5)$$

- Second Forward Difference - Differentiating the equation (1.1.3.5), we obtain:

$$y''_n = \frac{y'_{n+1} - y'_n}{h} \quad (1.1.3.6)$$

From equation (1.1.3.5), we obtain:

$$y'_{n+1} = \frac{y_{n+2} - y_{n+1}}{h}, y'_n = \frac{y_{n+1} - y_n}{h} \quad (1.1.3.7)$$

Substituting, we get:

$$y''_n = \frac{y_{n+2} - 2y_{n+1} + y_n}{h^2} \quad (1.1.3.8)$$

1.1.4 The Fourier Series - Any continuous or piecewise-continuous periodic function can be expressed as an infinite sum of complex exponentials.

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{j2\pi k f_0 t} \quad (1.1.4.1)$$

Multiply both sides by $e^{-j2\pi m f_0 t}$ (where m is an integer) and integrate over the interval $\left[-\frac{1}{2f_0}, \frac{1}{2f_0}\right]$

$$\int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x(t) e^{-j2\pi m f_0 t} dt \quad (1.1.4.2)$$

$$= \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} \left(\sum_{k=-\infty}^{\infty} c_k e^{j2\pi k f_0 t} \right) e^{-j2\pi m f_0 t} dt \quad (1.1.4.3)$$

$$= \sum_{k=-\infty}^{\infty} c_k \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} e^{j2\pi(k-m)f_0 t} dt \quad (1.1.4.4)$$

Using the orthogonality property over the period T_0

$$\int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} e^{j2\pi(k-m)f_0 t} dt = \begin{cases} \frac{1}{f_0}, & k = m \\ 0, & k \neq m \end{cases} \quad (1.1.4.5)$$

Substituting this back into the sum isolates the term for $k = m$:

$$\int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x(t) e^{-j2\pi m f_0 t} dt = c_m \cdot \frac{1}{f_0} \quad (1.1.4.6)$$

$$c_k = f_0 \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x(t) e^{-j2\pi k f_0 t} dt \quad (1.1.4.7)$$

1.1.5 Matrices

- Fundamental property:

$$|\mathbf{A}^\top| = |\mathbf{A}| \quad (1.1.5.1)$$

Solution: When expanding $|\mathbf{A}|$ along Row 1, you select entries across the top row and compute the determinants of their remaining submatrices. When expanding $|\mathbf{A}^\top|$ down Column 1, those exact same entries now sit vertically in the first column, and their corresponding submatrices are simply transposed.

- Matrix $\mathbf{A}$ (Expand along Row 1):

$$|\mathbf{A}| = \begin{pmatrix} \mathbf{a}_{11} & \mathbf{a}_{12} & \mathbf{a}_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \quad (1.1.5.2)$$

- Matrix $\mathbf{A}^\top$ (Expand down Column 1):

$$|\mathbf{A}^\top| = \begin{pmatrix} \mathbf{a}_{11} & a_{21} & a_{31} \\ \mathbf{a}_{12} & a_{22} & a_{32} \\ \mathbf{a}_{13} & a_{23} & a_{33} \end{pmatrix} \quad (1.1.5.3)$$

Submatrix Comparison for Entry a_{12} - When selecting a_{12} :

- In $\mathbf{A}$ (Row 1, Column 2 deletion):

$$\mathbf{M}_{12} = \begin{pmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{pmatrix} \quad (1.1.5.4)$$

– In $\mathbf{A}^\top$ (Row 2, Column 1 deletion):

$$\mathbf{N}_{21} = \begin{pmatrix} a_{21} & a_{31} \\ a_{23} & a_{33} \end{pmatrix} \quad (1.1.5.5)$$

Notice that:

$$\mathbf{N}_{21} = (\mathbf{M}_{12})^\top \quad (1.1.5.6)$$

Since:

$$|\mathbf{M}_{12}| = |(\mathbf{M}_{12})^\top| \quad (1.1.5.7)$$

Every term in the expansion of $|\mathbf{A}|$ is identically equal to the corresponding term in $|\mathbf{A}^\top|$.

- $|\mathbf{AB}| = |\mathbf{A}||\mathbf{B}|$

Solution: If $\mathbf{A}$ is invertible ($|\mathbf{A}| \neq 0$), it can be factored as a product of elementary matrices:

$$\mathbf{A} = \mathbf{E}_1 \mathbf{E}_2 \cdots \mathbf{E}_k \quad (1.1.5.8)$$

For any single elementary matrix $\mathbf{E}$, multiplying $\mathbf{EB}$ applies a single elementary row operation to $\mathbf{B}$:

$$|\mathbf{EB}| = |\mathbf{E}||\mathbf{B}| \quad (1.1.5.9)$$

Applying this property iteratively to the product $\mathbf{AB}$:

$$|\mathbf{AB}| = |\mathbf{E}_1 \mathbf{E}_2 \cdots \mathbf{E}_k \mathbf{B}| \quad (1.1.5.10)$$

$$= |\mathbf{E}_1| |\mathbf{E}_2 \cdots \mathbf{E}_k \mathbf{B}| \quad (1.1.5.11)$$

$$= |\mathbf{E}_1| |\mathbf{E}_2| \cdots |\mathbf{E}_k| |\mathbf{B}| \quad (1.1.5.12)$$

Setting $\mathbf{B}$ as the identity matrix $\mathbf{I}$:

$$|\mathbf{A}| = |\mathbf{E}_1| |\mathbf{E}_2| \cdots |\mathbf{E}_k| \quad (1.1.5.13)$$

Substituting $|\mathbf{A}|$ back into the expansion yields:

$$|\mathbf{AB}| = |\mathbf{A}||\mathbf{B}| \quad (1.1.5.14)$$

- Determinant of $\mathbf{A}^{-1}$, where $\mathbf{A}$ is an invertible square matrix

$$\mathbf{AA}^{-1} = \mathbf{I} \quad (1.1.5.15)$$

Taking the determinant of both sides:

$$|\mathbf{AA}^{-1}| = |\mathbf{I}| \quad (1.1.5.16)$$

From (1.1.5.14):

$$|\mathbf{A}||\mathbf{A}^{-1}| = 1 \quad (1.1.5.17)$$

Dividing both sides by $|\mathbf{A}|$ yields:

$$|\mathbf{A}^{-1}| = \frac{1}{|\mathbf{A}|} \quad (1.1.5.18)$$

- A few relations on the eigenvalues of a matrix - Let $\lambda_1, \lambda_2, \dots, \lambda_i$ be the eigenvalues of a matrix $\mathbf{A}$. Then, we have:

$$\sum_{i=1}^n \lambda_i = \text{Tr}(\mathbf{A}) \quad (1.1.5.19)$$

$$\prod_{i=1}^n \lambda_i = |\mathbf{A}| \quad (1.1.5.20)$$

- Eigenvalues of a square matrix $\mathbf{A}$ and its transpose $\mathbf{A}^\top$ - The characteristic polynomial of matrix $\mathbf{A}$ is defined as:

$$p_{\mathbf{A}}(\lambda) = |\mathbf{A} - \lambda \mathbf{I}| \quad (1.1.5.21)$$

From (1.1.5.1):

$$p_{\mathbf{A}}(\lambda) = |(\mathbf{A} - \lambda \mathbf{I})^\top| \quad (1.1.5.22)$$

$$= |\mathbf{A}^\top - (\lambda \mathbf{I})^\top| \quad (1.1.5.23)$$

$$= |\mathbf{A}^\top - \lambda \mathbf{I}| \quad (1.1.5.24)$$

$$= p_{\mathbf{A}^\top}(\lambda) \quad (1.1.5.25)$$

Since $\mathbf{A}$ and $\mathbf{A}^\top$ share the exact same characteristic polynomial, their eigenvalues are identical.

- Results on the conic sections
 - a) The equation of a conic with directrix $\mathbf{n}^\top \mathbf{x} = c$, eccentricity e and focus $\mathbf{F}$ is given by

$$g(\mathbf{x}) = \mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0 \quad (1.1.5.26)$$

where

$$\mathbf{V} = \|\mathbf{n}\|^2 \mathbf{I} - e^2 \mathbf{n} \mathbf{n}^\top, \quad (1.1.5.27)$$

$$\mathbf{u} = ce^2 \mathbf{n} - \|\mathbf{n}\|^2 \mathbf{F}, \quad (1.1.5.28)$$

$$f = \|\mathbf{n}\|^2 \|\mathbf{F}\|^2 - c^2 e^2 \quad (1.1.5.29)$$

Solution:

$$\|\mathbf{x} - \mathbf{F}\|^2 = e^2 \frac{(\mathbf{n}^\top \mathbf{x} - c)^2}{\|\mathbf{n}\|^2} \quad (1.1.5.30)$$

$$\Rightarrow \|\mathbf{n}\|^2 (\mathbf{x} - \mathbf{F})^\top (\mathbf{x} - \mathbf{F}) = e^2 (\mathbf{n}^\top \mathbf{x} - c)^2 \quad (1.1.5.31)$$

$$\Rightarrow \|\mathbf{n}\|^2 (\mathbf{x}^\top \mathbf{x} - 2\mathbf{F}^\top \mathbf{x} + \|\mathbf{F}\|^2) = e^2 (c^2 + (\mathbf{n}^\top \mathbf{x})^2 - 2c\mathbf{n}^\top \mathbf{x}) \quad (1.1.5.32)$$

$$= e^2 (c^2 + (\mathbf{x}^\top \mathbf{n} \mathbf{n}^\top \mathbf{x}) - 2c\mathbf{n}^\top \mathbf{x}) \quad (1.1.5.33)$$

which can be expressed as (1.1.5.26) after simplification.

b) The eccentricity of (1.1.5.26) is given by

$$e = \sqrt{1 - \frac{\lambda_1}{\lambda_2}} \quad (1.1.5.34)$$

Here, λ_1 and λ_2 are the eigenvalues of $\mathbf{V}$

Solution:

From (1.1.5.27), using the fact that $\mathbf{V}$ is symmetric,

$$\mathbf{V}^\top \mathbf{V} = \left(\|\mathbf{n}\|^2 \mathbf{I} - e^2 \mathbf{n} \mathbf{n}^\top \right)^\top \left(\|\mathbf{n}\|^2 \mathbf{I} - e^2 \mathbf{n} \mathbf{n}^\top \right) \quad (1.1.5.35)$$

$$\Rightarrow \mathbf{V}^2 = \|\mathbf{n}\|^4 \mathbf{I} + e^4 \mathbf{n} \mathbf{n}^\top \mathbf{n} \mathbf{n}^\top - 2e^2 \|\mathbf{n}\|^2 \mathbf{n} \mathbf{n}^\top \quad (1.1.5.36)$$

$$= \|\mathbf{n}\|^4 \mathbf{I} + e^4 \|\mathbf{n}\|^2 \mathbf{n} \mathbf{n}^\top - 2e^2 \|\mathbf{n}\|^2 \mathbf{n} \mathbf{n}^\top \quad (1.1.5.37)$$

$$= \|\mathbf{n}\|^4 \mathbf{I} + e^2 (e^2 - 2) \|\mathbf{n}\|^2 \mathbf{n} \mathbf{n}^\top \quad (1.1.5.38)$$

$$= \|\mathbf{n}\|^4 \mathbf{I} + (e^2 - 2) \|\mathbf{n}\|^2 (\|\mathbf{n}\|^2 \mathbf{I} - \mathbf{V}) \quad (1.1.5.39)$$

which can be expressed as

$$\mathbf{V}^2 + (e^2 - 2) \|\mathbf{n}\|^2 \mathbf{V} - (e^2 - 1) \|\mathbf{n}\|^4 \mathbf{I} = 0 \quad (1.1.5.40)$$

Using the Cayley-Hamilton theorem, (1.1.5.40) results in the characteristic equation,

$$\lambda^2 - (2 - e^2) \|\mathbf{n}\|^2 \lambda + (1 - e^2) \|\mathbf{n}\|^4 = 0 \quad (1.1.5.41)$$

which can be expressed as

$$\left(\frac{\lambda}{\|\mathbf{n}\|^2} \right)^2 - (2 - e^2) \left(\frac{\lambda}{\|\mathbf{n}\|^2} \right) + (1 - e^2) = 0 \quad (1.1.5.42)$$

$$\Rightarrow \frac{\lambda}{\|\mathbf{n}\|^2} = 1 - e^2, 1 \quad (1.1.5.43)$$

$$\text{or, } \lambda_2 = \|\mathbf{n}\|^2, \lambda_1 = (1 - e^2) \lambda_2 \quad (1.1.5.44)$$

c) (1.1.5.26) represents

- i) a parabola for $|\mathbf{V}| = 0$,
- ii) ellipse for $|\mathbf{V}| > 0$ and
- iii) hyperbola for $|\mathbf{V}| < 0$.

This is a direct result of the following two equations - From (1.1.5.34), we

have

$$\frac{\lambda_1}{\lambda_2} = 1 - e^2 \quad (1.1.5.45)$$

From (1.1.5.20), we have

$$|\mathbf{V}| = \lambda_1 \lambda_2 \quad (1.1.5.46)$$

- Matrix Determinant Lemma - Consider two column vectors $\mathbf{u}$ and $\mathbf{v}$

$$|\mathbf{I} + \mathbf{u}\mathbf{v}^\top| = 1 + \mathbf{v}^\top \mathbf{u} \quad (1.1.5.47)$$

1.1.6 SHM

- Springs in series: For n springs with spring constants $k_1, k_2, \dots, k_n$ connected in series under an applied tensile force F . - By Hooke's Law, the individual displacements are given by:

$$x_i = \frac{F}{k_i} \forall i \in \{1, 2, \dots, n\} \quad (1.1.6.1)$$

The total displacement in terms of the equivalent spring constant k_{eq} is:

$$x_{\text{total}} = \frac{F}{k_{\text{eq}}} \quad (1.1.6.2)$$

We know that

$$x_{\text{total}} = \sum_{i=1}^n x_i \quad (1.1.6.3)$$

Substituting $x_1, x_2, \dots, x_n$ and x_{total} into (1.1.6.3)

$$\frac{F}{k_{\text{eq}}} = \sum_{i=1}^n \frac{F}{k_i} \quad (1.1.6.4)$$

$$F \left(\frac{1}{k_{\text{eq}}} \right) = F \sum_{i=1}^n \left(\frac{1}{k_i} \right) \quad (1.1.6.5)$$

$$\frac{1}{k_{\text{eq}}} = \sum_{i=1}^n \frac{1}{k_i} \quad (1.1.6.6)$$

- General case of vertical SHM: Consider the following diagram. We need to find out the equation of SHM. When the mass M is at rest, let the system stretch by δ_{st} . Let w be the angular frequency of oscillation.

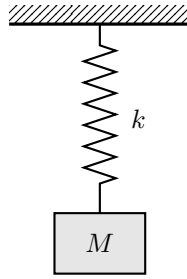


Fig. 1.1.6.1: Spring Mass system in vertical plane

$$\sum F = 0 \implies Mg - k\delta_{\text{st}} = 0 \implies Mg = k\delta_{\text{st}} \quad (1.1.6.7)$$

When the mass is displaced downward by $x(t)$, applying Newton's second law,

$$\sum F = Ma \quad (1.1.6.8)$$

$$\implies M \frac{d^2x}{dt^2} = Mg - k(\delta_{\text{st}} + x) \quad (1.1.6.9)$$

$$\implies M \frac{d^2x}{dt^2} = Mg - k\delta_{\text{st}} - kx \quad (1.1.6.10)$$

The static equilibrium condition is:

$$Mg = k\delta_{\text{st}} \quad (1.1.6.11)$$

Substituting this into (1.1.6.10)

$$M \frac{d^2x}{dt^2} = (k\delta_{\text{st}}) - k\delta_{\text{st}} - kx \quad (1.1.6.12)$$

$$M \frac{d^2x}{dt^2} = -kx \quad (1.1.6.13)$$

$$\frac{d^2x}{dt^2} = \frac{-k}{M}x = -w^2x \quad (1.1.6.14)$$

1.1.7 Results on Gravitation

- Specific angular momentum - Consider a satellite with position vector $\mathbf{r}$ and velocity vector $\mathbf{v}$

$$h_0 = |\mathbf{r} \times \mathbf{v}| \quad (1.1.7.1)$$

- Eccentricity of trajectory

$$e^2 = 1 + \frac{2\varepsilon h_0^2}{\mu^2} \quad (1.1.7.2)$$

- Trajectory of a satellite given its position vector, velocity vector, and gravitational parameter μ : The polar trajectory of a satellite with eccentricity vector $\mathbf{e}$ and semi-latus rectum p_{orb} is given by:

$$r = \frac{p_{\text{orb}}}{1 + e \cos \theta} \quad (1.1.7.3)$$

In vector form:

$$e \cos \theta = \frac{\mathbf{e}^\top \mathbf{x}}{r} \quad (1.1.7.4)$$

Where:

$$\mathbf{x} = \begin{pmatrix} r \cos \theta \\ r \sin \theta \end{pmatrix}. \quad (1.1.7.5)$$

From (1.1.7.3) and (1.1.7.4), we have:

$$r + \mathbf{e}^\top \mathbf{x} = p_{\text{orb}} \implies r = p_{\text{orb}} - \mathbf{e}^\top \mathbf{x} \quad (1.1.7.6)$$

Squaring both sides :

$$r^2 = \mathbf{x}^\top \mathbf{x} = (p_{\text{orb}} - \mathbf{e}^\top \mathbf{x})^2 = p_{\text{orb}}^2 - 2p_{\text{orb}}\mathbf{e}^\top \mathbf{x} + (\mathbf{e}^\top \mathbf{x})^2 \quad (1.1.7.7)$$

Rearranging yields:

$$\mathbf{x}^\top (\mathbf{I} - \mathbf{e}\mathbf{e}^\top) \mathbf{x} + 2(p_{\text{orb}}\mathbf{e})^\top \mathbf{x} - p_{\text{orb}}^2 = 0 \quad (1.1.7.8)$$

Comparing this with (1.1.5.26):

- Matrix $\mathbf{V}$: $\mathbf{V} = \mathbf{I} - \mathbf{e}\mathbf{e}^\top$
- Vector $\mathbf{u}$: $\mathbf{u} = p_{\text{orb}}\mathbf{e}$
- Scalar f : $f = -p_{\text{orb}}^2$

From result (1.1.5.47), we have:

$$|\mathbf{V}| = 1 - \mathbf{e}^\top \mathbf{e} = 1 - e^2 \quad (1.1.7.9)$$

From the sign of $|\mathbf{V}|$, we get the trajectory.

B. Problems

1.2.1 How many 3-digit numbers can be formed using three distinct single digit prime numbers? (GATE AE 2026)

- 64
- 24
- 12
- 4

Solution: There are 4 single digit prime numbers: 2,3,5 and 7. The desired numbers are given by

$${}^4P_3 = \frac{4!}{1!} = 24 \quad (1.2.1.1)$$

1.2.2 In a group of students, 10 students like Mathematics, 12 students like English, 4 students like both Mathematics and English, and 6 students like neither Mathematics nor English. The number of students in the group is _____. (GATE AE 2026)

- a) 18
- b) 20
- c) 24
- d) 32

Solution: Let N be the total number of students in the group. If one student is chosen uniformly at random from the group, we define:

- M : The event that the student likes Mathematics.
- E : The event that the student likes English.

The respective probabilities are expressed as:

$$P(M) = \frac{10}{N}, \quad P(E) = \frac{12}{N} \quad (1.2.2.1)$$

$$P(ME) = \frac{4}{N} \quad (\text{student likes both}) \quad (1.2.2.2)$$

$$P((M + E)') = \frac{6}{N} \quad (\text{student likes neither}) \quad (1.2.2.3)$$

From (1.1.1.3)

$$P(M + E) + P((M + E)') = 1 \quad (1.2.2.4)$$

From (1.1.1.1)

$$P(M) + P(E) - P(ME) + P((M + E)') = 1 \quad (1.2.2.5)$$

Substitute the probabilities in terms of N :

$$\frac{10}{N} + \frac{12}{N} - \frac{4}{N} + \frac{6}{N} = 1 \quad (1.2.2.6)$$

$$\frac{10 + 12 - 4 + 6}{N} = 1 \quad (1.2.2.7)$$

$$\frac{24}{N} = 1 \implies N = 24 \quad (1.2.2.8)$$

1.2.3 Given a square of side $s = 4\text{cm}$ and two circles, one with known radius $r_1 = 1\text{cm}$ and another with unknown radius r_2 , such that both of them touch each other externally and also the square, find the value of r_2 . Also describe how the figure in the problem can be constructed. (GATE AE 2026)

Circle C_2 (r_2):

$$\mathbf{O}_2 = \begin{pmatrix} s - r_2 \\ s - r_2 \end{pmatrix}, \quad \mathbf{S} = \begin{pmatrix} s - r_2 \\ s \end{pmatrix}, \quad \mathbf{R} = \begin{pmatrix} s \\ s - r_2 \end{pmatrix} \quad (1.2.3.7)$$

Line of centers and contact point $\mathbf{T}$:

$$\mathbf{O}_2 - \mathbf{O}_1 = \begin{pmatrix} s - r_2 - r_1 \\ s - r_2 - r_1 \end{pmatrix} = (s - r_2 - r_1) \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (1.2.3.8)$$

Now, given that the circles are touching, we know that:

$$\frac{O_1 T}{O_2 T} = \frac{r_1}{r_2} \quad (1.2.3.9)$$

$$\frac{O_1 T}{O_1 O_2} = \frac{r_1}{r_1 + r_2} \quad (1.2.3.10)$$

Hence, using (1.2.3.10) in (1.2.3.8) gives us:

$$\mathbf{T} = \mathbf{O}_1 + \frac{r_1}{r_1 + r_2} (\mathbf{O}_2 - \mathbf{O}_1) \quad (1.2.3.11)$$

The following code simulates the construction.

codes/Circle.py

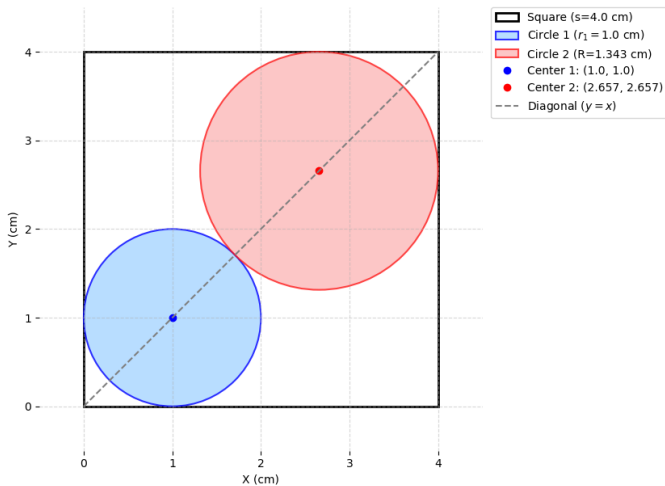


Fig. 1.2.3.1: Square with Tangent Circles

1.2.4 Consider the contour C shown in the figure below. For the vector $\mathbf{F} = (x + 2y) \hat{e}_x + (2x + 4y) \hat{e}_y$, the integral $\oint_C \mathbf{F} \cdot d\mathbf{l} = \underline{\hspace{2cm}}$.

Here $d\mathbf{l}$ represents an infinitesimal length along the contour C . (GATE AE 2026)

a) 0

- b) 2
c) 4
d) 6

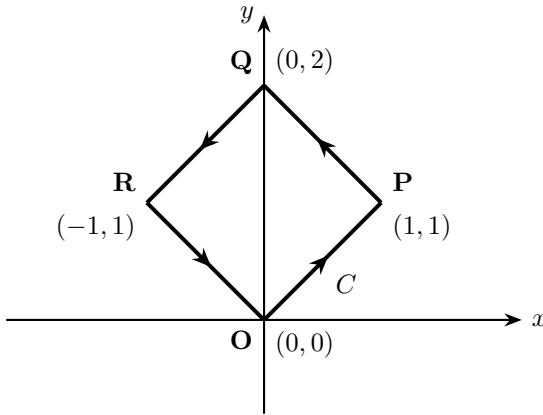


Fig. 1.2.4.1: Contour C

Solution: We can break the closed path C into its four straight line paths: $O \rightarrow P \rightarrow Q \rightarrow R \rightarrow O$.

$$\mathbf{F} \cdot d\mathbf{l} = \begin{pmatrix} x + 2y \\ 2x + 4y \end{pmatrix} \cdot \begin{pmatrix} dx \\ dy \end{pmatrix} \quad (1.2.4.1)$$

$$\mathbf{F}^\top d\mathbf{l} = \begin{pmatrix} x + 2y \\ 2x + 4y \end{pmatrix}^\top \begin{pmatrix} dx \\ dy \end{pmatrix} = (x + 2y \quad 2x + 4y) \begin{pmatrix} dx \\ dy \end{pmatrix} \quad (1.2.4.2)$$

$$= ((x + 2y)dx + (2x + 4y)dy) \quad (1.2.4.3)$$

The general line integral expression to evaluate along any path is:

$$I = \int (x + 2y) dx + (2x + 4y) dy \quad (1.2.4.4)$$

Path $O(0,0) \rightarrow P(1,1)$:

- Equation of line:

$$y = x \implies dy = dx \quad (1.2.4.5)$$

- Limits: x goes from 0 to 1

$$I_1 = \int_0^1 (x + 2x) dx + (2x + 4x) dx = \int_0^1 9x dx = \left[\frac{9x^2}{2} \right]_0^1 = \frac{9}{2} \quad (1.2.4.6)$$

Path $P(1,1) \rightarrow Q(0,2)$:

- Equation of line:

$$y = 2 - x \implies dy = -dx \quad (1.2.4.7)$$

- Limits: x goes from 1 to 0

$$I_2 = \int_1^0 (x + 2(2 - x)) dx + (2x + 4(2 - x))(-dx) \quad (1.2.4.8)$$

$$I_2 = \int_1^0 [(4 - x) - (8 - 2x)] dx = \int_1^0 (x - 4) dx = \left[\frac{x^2}{2} - 4x \right]_1^0 = \frac{7}{2} \quad (1.2.4.9)$$

Path $Q(0, 2) \rightarrow R(-1, 1)$:

- Equation of line:

$$y = x + 2 \implies dy = dx \quad (1.2.4.10)$$

- Limits: x goes from 0 to -1

$$I_3 = \int_0^{-1} (x + 2(x + 2)) dx + (2x + 4(x + 2)) dx = \int_0^{-1} (9x + 12) dx \quad (1.2.4.11)$$

$$I_3 = \left[\frac{9x^2}{2} + 12x \right]_0^{-1} = \left(\frac{9}{2} - 12 \right) - 0 = -\frac{15}{2} \quad (1.2.4.12)$$

Path $R(-1, 1) \rightarrow O(0, 0)$:

- Equation of line:

$$y = -x \implies dy = -dx \quad (1.2.4.13)$$

- Limits: x goes from -1 to 0

$$I_4 = \int_{-1}^0 (x + 2(-x)) dx + (2x + 4(-x))(-dx) = \int_{-1}^0 x dx \quad (1.2.4.14)$$

$$I_4 = \left[\frac{x^2}{2} \right]_{-1}^0 = 0 - \frac{1}{2} = -\frac{1}{2} \quad (1.2.4.15)$$

Summing the integrals across all four line segments:

$$I = I_1 + I_2 + I_3 + I_4 \quad (1.2.4.16)$$

$$I = \frac{9}{2} + \frac{7}{2} - \frac{15}{2} - \frac{1}{2} = \frac{16 - 16}{2} = 0 \quad (1.2.4.17)$$

The following code simulates the result.

codes/Line_Integral.py

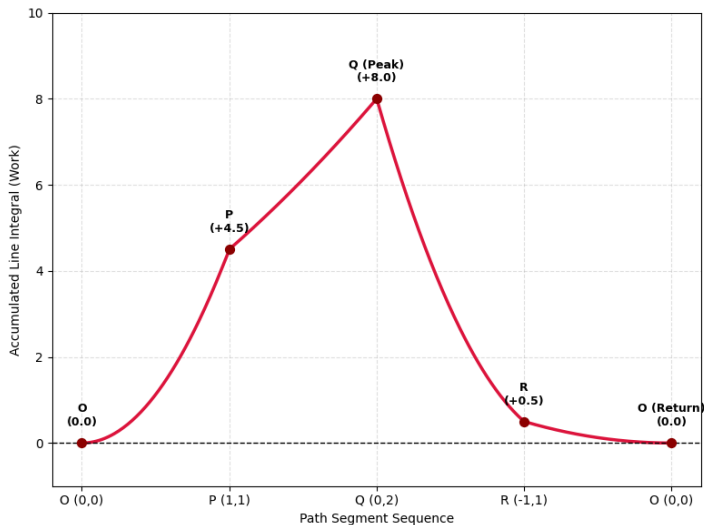


Fig. 1.2.4.2: Visualising the Cancellation of Integrals

1.2.5 The Fourier series representation of a square wave is shown in the figure below. The fluctuations seen near $x = \pm 1$ are named after which one of the following scientists? (GATE AE 2026)

- a) Cauchy
- b) Fourier
- c) Gibbs
- d) Laplace

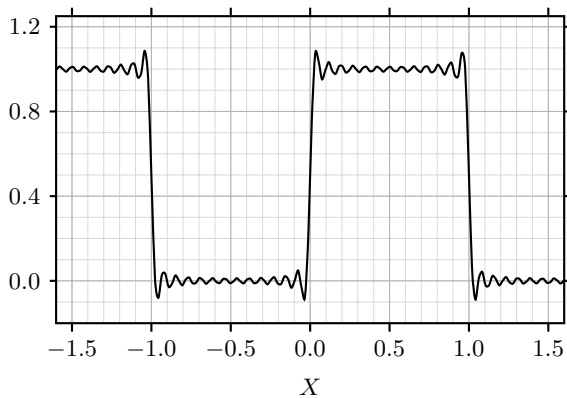


Fig. 1.2.5.1: Fourier Square Wave Representation

Solution:

- a) Gibbs Phenomenon

- Explanation of Gibbs Phenomenon: Every periodic function can be expressed as a sum of sines and cosines. Gibbs phenomenon is the overshoot and high-frequency oscillation ("ringing") that occurs when approximating a function with jump discontinuities using a finite (truncated) Fourier series. Even as the number of terms N approaches infinity, the overshoot near the jump discontinuity does not vanish; instead, it narrows in width while maintaining a fixed percentage height relative to the size of the jump.
 - Reason: When approximating a step function or square wave, Fourier series synthesize the shape using smooth, continuous sinusoids (sin and cos). Because continuous functions cannot instantaneously step from one value to another, the Fourier series overcompensates near the vertical edge.
- b) Finding the fourier coefficients for a square wave defined as-

$$u(t) = \begin{cases} 1, & \text{if } t > 0 \\ \frac{1}{2}, & \text{if } t = 0 \\ 0, & \text{if } t < 0 \end{cases} \quad (1.2.5.1)$$

Consider:

$$f_0 = \frac{1}{4} \quad (1.2.5.2)$$

We will use the result (1.1.4.7). Considering the interval $\left(-\frac{1}{2f_0}, \frac{1}{2f_0}\right)$

$$c_k = \frac{1}{4} \int_{-2}^2 u(t) e^{-j \frac{\pi k t}{2}} dt \quad (1.2.5.3)$$

From (1.2.5.1), we have

$$c_k = \frac{1}{4} \int_{-1}^1 e^{-j \frac{\pi k t}{2}} dt \quad (1.2.5.4)$$

$$c_k = \begin{cases} \frac{1}{4} \int_{-1}^1 1 dt = \frac{1}{2}, & k = 0 \\ \frac{1}{4} \left[\frac{e^{-j \frac{\pi k t}{2}}}{-j \frac{\pi k}{2}} \right]_{-1}^1 = \frac{1}{2} \operatorname{sinc} \left(\frac{\pi k}{2} \right), & k \neq 0 \end{cases} \quad (1.2.5.5)$$

The following code verifies the result.

codes/Fourier.py

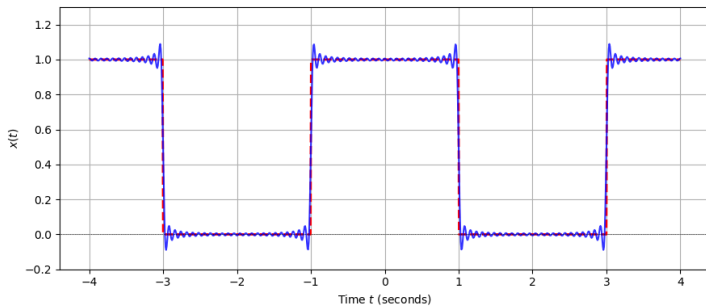


Fig. 1.2.5.2: Verification With Plot

1.2.6 If a matrix can be written as $\mathbf{A} = \mathbf{uv}^\top$, where both u and v are n -dimensional real-valued non-zero column vectors, then the rank of the matrix $\mathbf{A}$ is? (GATE AE 2026)

Solution: We will solve the above problem with an example. Consider 3-dimensional non-zero vectors:

$$\mathbf{u} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \mathbf{v} = \begin{pmatrix} 1 \\ 4 \\ 2 \end{pmatrix} \quad (1.2.6.1)$$

The matrix $\mathbf{A}$ is formed as:

$$\mathbf{A} = \mathbf{uv}^\top = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} (1 \quad 4 \quad 2) \quad (1.2.6.2)$$

$$= \begin{pmatrix} 1 & 4 & 2 \\ 2 & 8 & 4 \\ 3 & 12 & 6 \end{pmatrix} \quad (1.2.6.3)$$

We will now apply elementary row transformations step by step. First we eliminate the non-zero entry in the second row using

$$\mathbf{A} \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \begin{pmatrix} 1 & 4 & 2 \\ 0 & 0 & 0 \\ 3 & 12 & 6 \end{pmatrix} \quad (1.2.6.4)$$

Now we eliminate the non-zero entry in the third row.

$$\xrightarrow{R_3 \rightarrow R_3 - 3R_1} \begin{pmatrix} 1 & 4 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad (1.2.6.5)$$

The matrix is now in row echelon form with only 1 non-zero row. Hence,

$$\text{rank}(\mathbf{A}) = 1 \quad (1.2.6.6)$$

1.2.7 An $n \times n$ square matrix $\mathbf{A}$ satisfies $\mathbf{A}^\top = \mathbf{A}^{-1}$. The determinant of this matrix may take which of the following value(s)? (GATE AE 2026)

- a) 1
- b) -1
- c) n
- d) 0

Solution: We are given that matrix A satisfies:

$$\mathbf{A}^\top = \mathbf{A}^{-1} \quad (1.2.7.1)$$

As the inverse exists, we know that

$$|\mathbf{A}| \neq 0 \quad (1.2.7.2)$$

Taking the determinant of both sides of the given matrix equation:

$$|\mathbf{A}^\top| = |\mathbf{A}^{-1}| \quad (1.2.7.3)$$

Substituting results (1.1.5.1) and (1.1.5.18) in (1.2.7.3), we get:

$$|\mathbf{A}| = \frac{1}{|\mathbf{A}|} \quad (1.2.7.4)$$

Multiplying both sides by $|\mathbf{A}|$:

$$(|\mathbf{A}|)^2 = 1 \implies |\mathbf{A}| = \pm 1 \quad (1.2.7.5)$$

1.2.8 Given the velocity and position vectors of a satellite along with its gravitational parameter $\mu = GM$, determine the trajectory of the satellite. (GATE AE 2026)

- a) Circle
- b) Hyperbola
- c) Parabola
- d) Straight Line

Position vector:

$$\mathbf{r} = 8000\hat{p} + 9000\hat{q} \text{ km} \quad (1.2.8.1)$$

Velocity vector:

$$\mathbf{v} = -6\hat{p} + 6\hat{q} \text{ km/s} \quad (1.2.8.2)$$

Gravitational parameter:

$$\mu = 398600 \text{ km}^3/\text{s}^2 \quad (1.2.8.3)$$

Solution: We are given :

$$\mathbf{r} = \begin{pmatrix} 8000 \\ 9000 \end{pmatrix}, \mathbf{v} = \begin{pmatrix} -6 \\ 6 \end{pmatrix} \quad (1.2.8.4)$$

From (1.1.7.1), Specific Angular Momentum h_0

$$= |8000(6) - 9000(-6)| = 102000 \text{ km}^2/\text{s} \quad (1.2.8.5)$$

From (1.1.7.2), eccentricity squared e^2

$$= 1 + \frac{2(2.898)(102000)^2}{(398600)^2} \approx 1 + 0.3795 = 1.3795 \quad (1.2.8.6)$$

From (1.1.7.9), we have:

$$|\mathbf{V}| = 1 - 1.3795 = -0.3795 \quad (1.2.8.7)$$

From 1.1.5(c)iii, we can conclude that the trajectory of the satellite is a Hyperbola. The following code simulates the result.

codes/Satellite.py

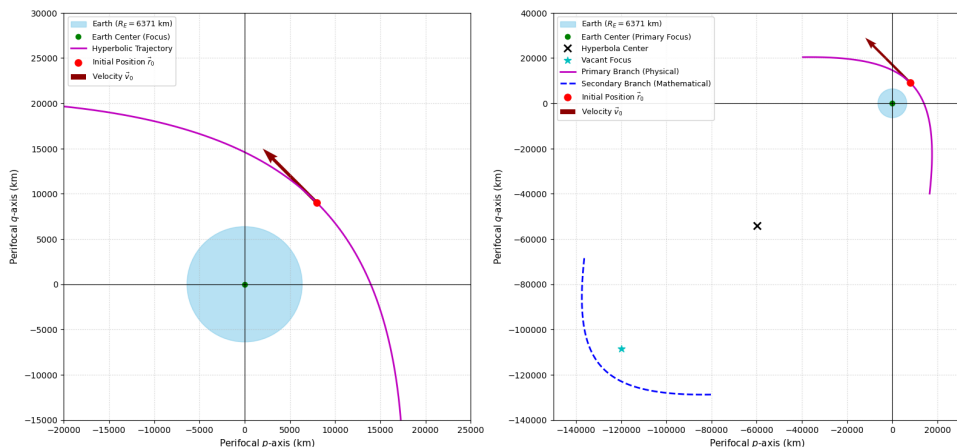


Fig. 1.2.8.1: Trajectory of the Satellite

1.2.9 For the matrix $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, if the relation $a + b = c + d$ holds and $a, b, c, d \neq 0$, then which one of the following statements about A is FALSE? (GATE AE 2026)

- $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ is an eigenvector
- $\lambda = a + b$ is an eigenvalue
- $\lambda = d - b$ is an eigenvalue
- $\lambda = d + b$ is an eigenvalue

Solution: Given:

$$a + b = c + d = k \quad (1.2.9.1)$$

Writing this in vector form:

$$(a \ b) \begin{pmatrix} 1 \\ 1 \end{pmatrix} = k = (c \ d) \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (1.2.9.2)$$

$$\Rightarrow \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} k \\ k \end{pmatrix} \quad (1.2.9.3)$$

$$\Rightarrow \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = k \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (1.2.9.4)$$

$\Rightarrow \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ is an eigenvector and $a + b$ is an eigenvalue.

From (1.2.9.1),

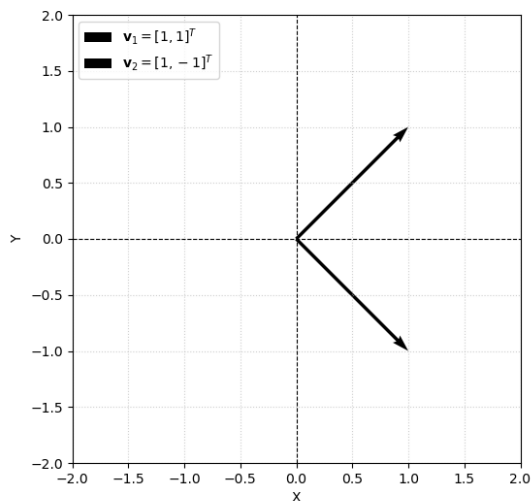
$$a - c = d - b = -(b - d) = l \quad (1.2.9.5)$$

Evaluating the matrix-vector multiplication:

$$\begin{pmatrix} a & c \\ b & d \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} a - c \\ b - d \end{pmatrix} = \begin{pmatrix} l \\ -l \end{pmatrix} = l \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad (1.2.9.6)$$

From (1.1.5.25), $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ is an eigenvector and $d - b$ is the 2nd eigenvalue. The following code simulates the result by plotting the eigenvectors.

codes/Eigenvectors.py



1.2.10 Consider the differential equation:

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 0 \quad (1.2.10.1)$$

with initial conditions $y|_{x=0} = 0$ and $\left.\frac{dy}{dx}\right|_{x=0} = 1$. Find the value of the slope $\frac{dy}{dx}$ at $x = \ln(2)$ rounded off to three decimal places. (GATE AE 2026)

Solution:

a) Theoretical solution - Let

$$y(x) \xrightarrow{\mathcal{L}} Y(s) \quad (1.2.10.2)$$

Taking the Laplace transform on both sides of (1.2.10.1)

$$\mathcal{L}\left\{\frac{d^2y}{dx^2}\right\} + 2\mathcal{L}\left\{\frac{dy}{dx}\right\} + \mathcal{L}\{y\} = 0 \quad (1.2.10.3)$$

From equations (1.1.2.6) and (1.1.2.12):

$$[s^2Y(s) - sy(0) - y'(0)] + 2[sY(s) - y(0)] + Y(s) = 0 \quad (1.2.10.4)$$

Substituting the given initial conditions:

$$y(0) = 0, y'(0) = 1 \quad (1.2.10.5)$$

$$(s^2Y(s) - 1) + 2sY(s) + Y(s) = 0 \quad (1.2.10.6)$$

$$(s^2 + 2s + 1)Y(s) = 1 \quad (1.2.10.7)$$

$$Y(s) = \frac{1}{(s+1)^2} \quad (1.2.10.8)$$

Taking the inverse Laplace transform by applying the first frequency shift theorem((1.1.2.14)) on (1.1.2.21) gives the solution $y(x)$:

$$y(x) = xe^{-x} \quad (1.2.10.9)$$

Differentiating $y(x)$ with respect to x :

$$\frac{dy}{dx} = \frac{d}{dx}(xe^{-x}) = (1-x)e^{-x} \quad (1.2.10.10)$$

$$y'(\ln(2)) = (1 - \ln(2))e^{-\ln(2)} \quad (1.2.10.11)$$

$$= \frac{1 - \ln(2)}{2} \approx \frac{1 - 0.693147}{2} \approx 0.153 \quad (1.2.10.12)$$

b) Numerical Solution - Substituting equations (1.1.3.5) and (1.1.3.8) into our

differential equation, we obtain:

$$\frac{y_{n+2} - 2y_{n+1} + y_n}{h^2} + 2 \left(\frac{y_{n+1} - y_n}{h} \right) + y_n = 0 \quad (1.2.10.13)$$

Rearranging, we get:

$$y_{n+2} = 2(1 - h)y_{n+1} - (1 - h)^2 y_n \quad (1.2.10.14)$$

Our initial conditions are:

$$y_0 = y(0) = 0, y_1 = y_0 + h(y'(0)) \quad (1.2.10.15)$$

We assume the following:

$$ln2 = 0.7, h = 0.01 \quad (1.2.10.16)$$

Then:

$$y_1 = 0 + 0.01(1) = 0.01 \quad (1.2.10.17)$$

Now, we continue this process until we find y_{69} and y_{70}

$$y_{69} = 0.3483, y_{70} = 0.3499 \quad (1.2.10.18)$$

From (1.1.3.5), we get:

$$y'_{70} = \frac{y_{70} - y_{69}}{h} \approx 0.145 \quad (1.2.10.19)$$

The following code simulates the difference between the numerical and theoretical solutions.

codes/Euler.py

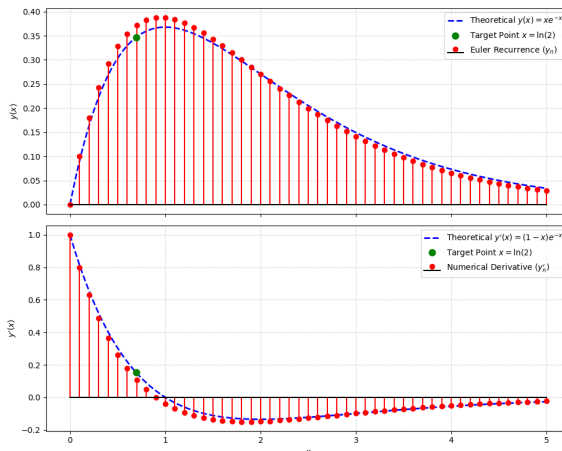


Fig. 1.2.10.1: Theoretical Plot vs Plot obtained with Euler's Forward Difference Method

1.2.11 Find the minimum value of the function $f(x) = |x| + |2x + 3|$ for real x . (GATE AE 2026)

Solution: The critical points occur where the expressions inside the absolute values equal zero:

$$x = 0 \quad (1.2.11.1)$$

$$2x + 3 = 0 \implies x = -\frac{3}{2} = -1.5 \quad (1.2.11.2)$$

These two critical points divide the real line into three sub-intervals:

$$(-\infty, -1.5), \quad [-1.5, 0), \quad \text{and} \quad [0, \infty) \quad (1.2.11.3)$$

- For $x < -1.5$, both expressions inside the absolute values are negative:

$$f(x) = -x - (2x + 3) = -3x - 3 \quad (1.2.11.4)$$

- For $-1.5 \leq x < 0$, the term x is negative while $2x + 3$ is non-negative:

$$f(x) = -x + (2x + 3) = x + 3 \quad (1.2.11.5)$$

- For $x \geq 0$, both expressions are non-negative:

$$f(x) = x + (2x + 3) = 3x + 3 \quad (1.2.11.6)$$

Combining these cases gives the complete piecewise definition:

$$f(x) = \begin{cases} -3x - 3, & \text{if } x < -1.5 \\ x + 3, & \text{if } -1.5 \leq x < 0 \\ 3x + 3, & \text{if } x \geq 0 \end{cases} \quad (1.2.11.7)$$

Analyzing the behavior of each linear segment:

- On $(-\infty, -1.5)$, the line $f(x)$ has a negative slope (-3) , so $f(x)$ strictly decreases as x approaches -1.5 .
- On $[-1.5, 0)$, the line $f(x)$ has a positive slope $(+1)$, so $f(x)$ strictly increases as x moves away from -1.5 .
- On $[0, \infty)$, the line $f(x)$ has a positive slope $(+3)$, so $f(x)$ continues to increase.

Since $f(x)$ decreases for $x < -1.5$ and increases for $x > -1.5$, the minimum value occurs at $x = -1.5$:

$$f(-1.5) = -1.5 + 3 = 1.5 \quad (1.2.11.8)$$

The following code plots the function.

codes/Modulus_function.py

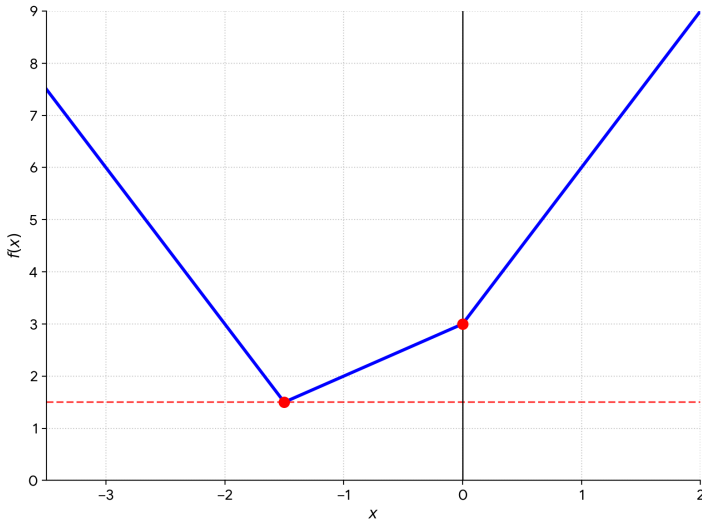


Fig. 1.2.11.1: Plot of the function

1.2.12 A bar, spring, and mass system is arranged in series.

- Bar: $E = 200 \text{ GPa}$, $A = 100 \text{ mm}^2$, $L = 100 \text{ mm}$
- Spring Stiffness: $k = 200 \text{ kN/mm}$
- Mass: $M = 100 \text{ kg}$

Find the natural frequency of free vibration ω_n in rad/s. Also solve the differential equation obtained to obtain the position, velocity and acceleration of the system. (GATE AE 2026)

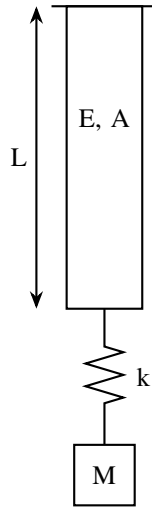


Fig. 1.2.12.1: Diagram of the system

Solution: The axial stiffness of the uniform bar is:

$$k_{\text{bar}} = \frac{AE}{L} \quad (1.2.12.1)$$

$$k_{\text{bar}} = \frac{10^{-4} \times 200 \times 10^9}{0.1} = 2 \times 10^8 \text{ N/m} \quad (1.2.12.2)$$

$$k = 200 \text{ kN/mm} = 2 \times 10^8 \text{ N/m} \quad (1.2.12.3)$$

Substituting the values of k and k_{eq} in (1.1.6.6) gives us

$$k_{\text{eq}} = \frac{k_{\text{bar}} \cdot k}{k_{\text{bar}} + k} = \frac{(2 \times 10^8) (2 \times 10^8)}{2 \times 10^8 + 2 \times 10^8} = 10^8 \text{ N/m} \quad (1.2.12.4)$$

The natural frequency of free vibration is:

$$\omega_n = \sqrt{\frac{k_{\text{eq}}}{M}} = \sqrt{\frac{10^8}{100}} = \sqrt{10^6} = 1000 \text{ rad/s} \quad (1.2.12.5)$$

Now, to find the position of the system as a function of time, from (1.1.6.14), we have:

$$\frac{d^2x}{dt^2} = \frac{-k}{M}x = -\omega_n^2 x \quad (1.2.12.6)$$

Let the initial displacement of the system be x_0 and initial velocity be v_0 . From result (1.1.2.37), we get:

$$x(t) = x_0 \cos(\omega_n t) + \frac{v_0}{\omega_n} \sin(\omega_n t) \quad (1.2.12.7)$$

Substituting the value of w_n ,

$$x(t) = x_0 \cos(1000t) + \frac{v_0}{1000} \sin(1000t) \quad (1.2.12.8)$$

Differentiating to get velocity:

$$v(t) = -1000x_0 \sin(1000t) + v_0 \cos(1000t) \quad (1.2.12.9)$$

Differentiating again to get the acceleration:

$$a(t) = -10^6 x_0 \cos(1000t) - 1000v_0 \sin(1000t) \quad (1.2.12.10)$$

The following code simulates the result.

codes/Spring.py

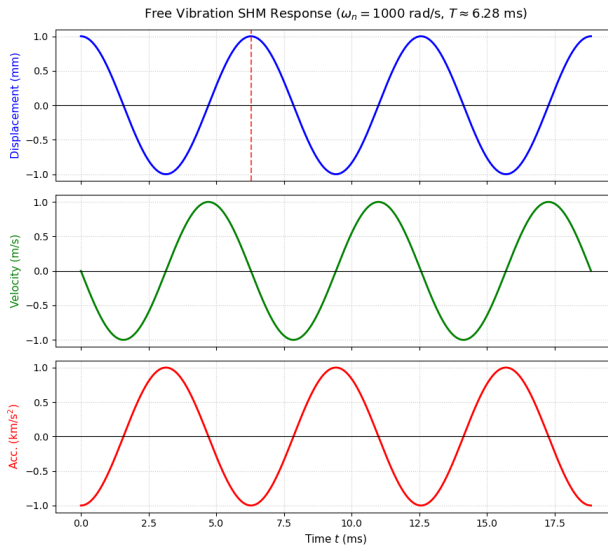


Fig. 1.2.12.2: Acceleration, Velocity and Position of the System